DATA ANALYTICS CAPSTONE

Banking Customer Churn Analysis

Financial Services · Dataset: 28,382 Customers · Python · Tableau

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PROBLEM STATEMENT

Why Churn Is a Critical Business Problem

Customer churn directly erodes bank revenue, LTV, and growth potential. Retaining existing customers is significantly more cost-effective than acquiring new ones — making churn prediction a strategic priority.

Revenue Impact

Every churned customer represents lost deposits, fees, and cross-sell revenue — compounding over time.

Retention vs. Acquisition

Retaining a customer costs a fraction of acquiring a new one. Reducing churn is the highest-ROI lever available.

Our Objective

Identify the key drivers of churn using EDA and statistical analysis, and surface actionable retention strategies.

DATA ENGINEERING

From Raw Data to Analysis-Ready Dataset

Source Dataset

*Banking customer dataset containing **28,382 customer records** across demographics, account behavior, and activity levels.*

Tools Used

- *Python — Pandas, NumPy, Seaborn*
- *Exploratory Data Analysis (EDA)*
- *Statistical Analysis*
- *Tableau (Visualization)*

01

Remove Duplicates

Identified and dropped duplicate records to ensure data integrity.

02

Handle Missing Values

Imputed or removed null entries using domain-appropriate strategies.

03

Encode Categorical Variables

Converted text fields (gender, geography) into model-ready formats.

04

Create Segments

*Engineered **Age Groups**, **Balance Segments**, and **Activity Levels** for richer analysis.*

KPI FRAMEWORK

Core Business Metrics at a Glance

These four numbers define the scale of the churn problem and frame every analysis that follows.

28,382

Total Customers

Full dataset used for analysis

5,260

Total Churned

Customers who left the bank

23,122

Retained

Customers still active

18.53%

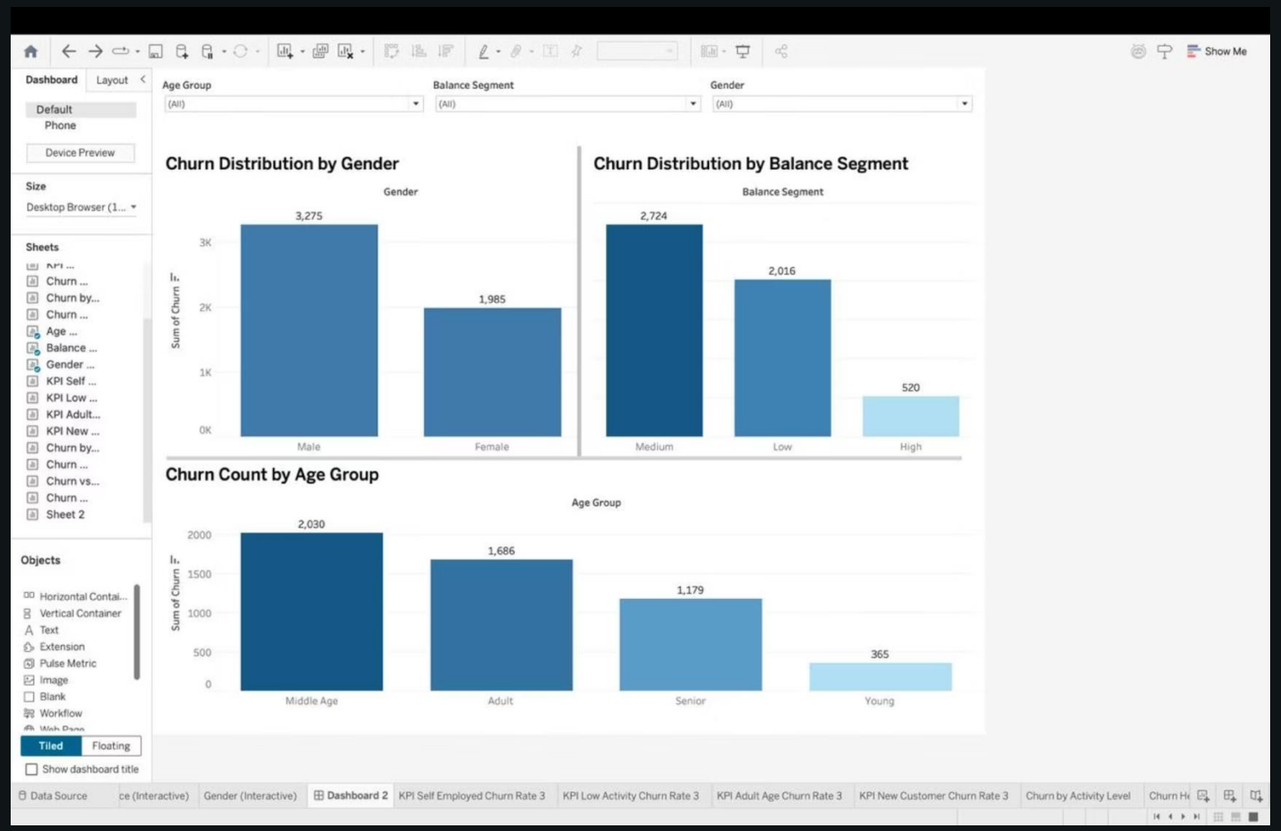
Churn Rate

Nearly 1 in 5 customers churned

⚠ *An 18.53% churn rate in financial services is significantly above industry benchmarks — representing a material risk to revenue and customer lifetime value that demands immediate strategic action.*

EDA INSIGHTS — SLIDE 5

Key Patterns Across Demographics & Behavior



Churn distribution by Gender, Balance Segment, and Age Group — Tableau EDA Dashboard

Activity Drives Churn

*Activity level is the single strongest predictor — but the direction is **counterintuitive**.*

Age Group Variation

Senior and Adult age brackets show disproportionate churn contribution by volume.

Balance Segment Split

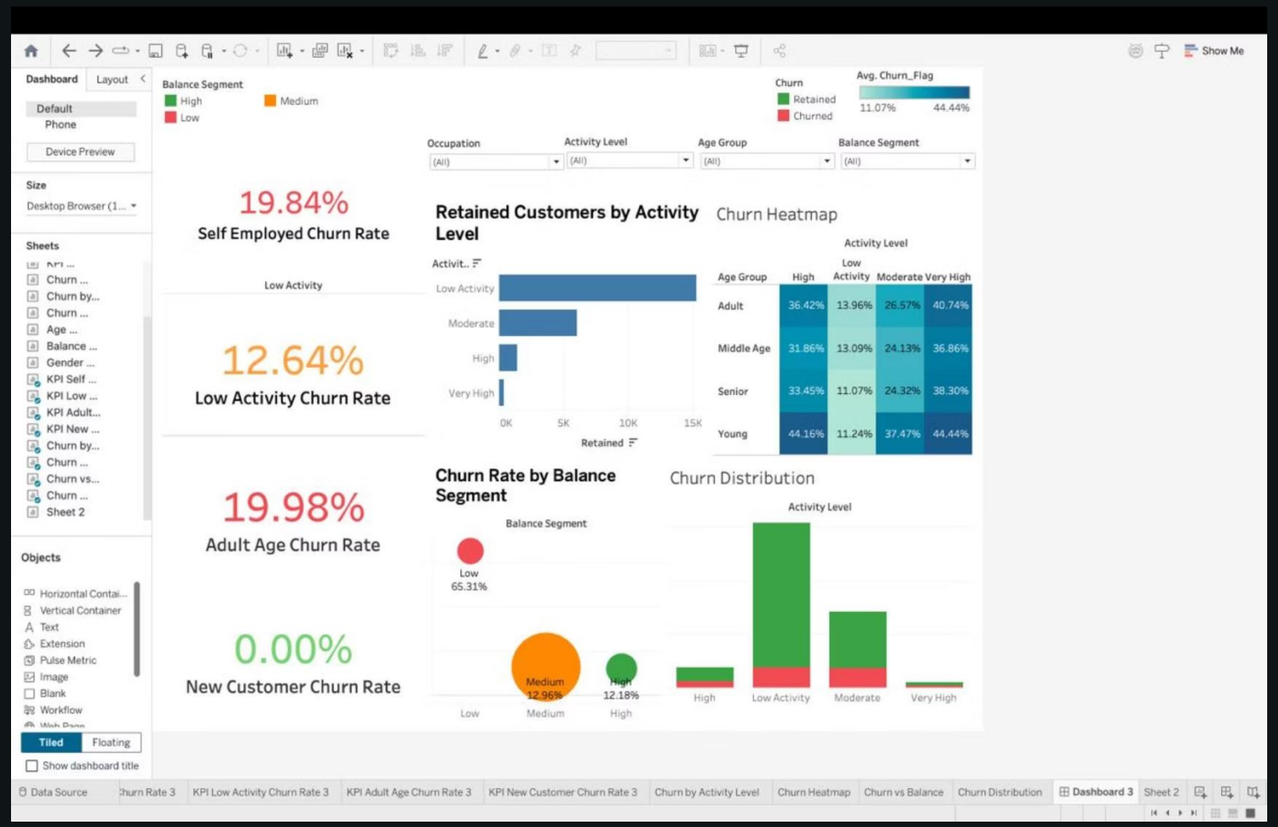
Mid-range balance customers contribute the majority of churned accounts.

Gender Distribution

Churn is distributed across genders, with female customers churning at a slightly elevated rate.

ADVANCED ANALYSIS — SLIDE 6

The Counterintuitive Finding: High Activity = High Churn



Churn Heat Map, Activity-Level Retention, KPI Cards & Balance Segment Breakdown

High engagement ≠ loyalty. *The most active customers are churning at the highest rates — signaling dissatisfaction or unmet expectations, not disengagement.*

38.37%

Very High Activity

Highest churn rate — most alarming segment

33.96%

High Activity

Second highest — reinforces the paradox

12.64%

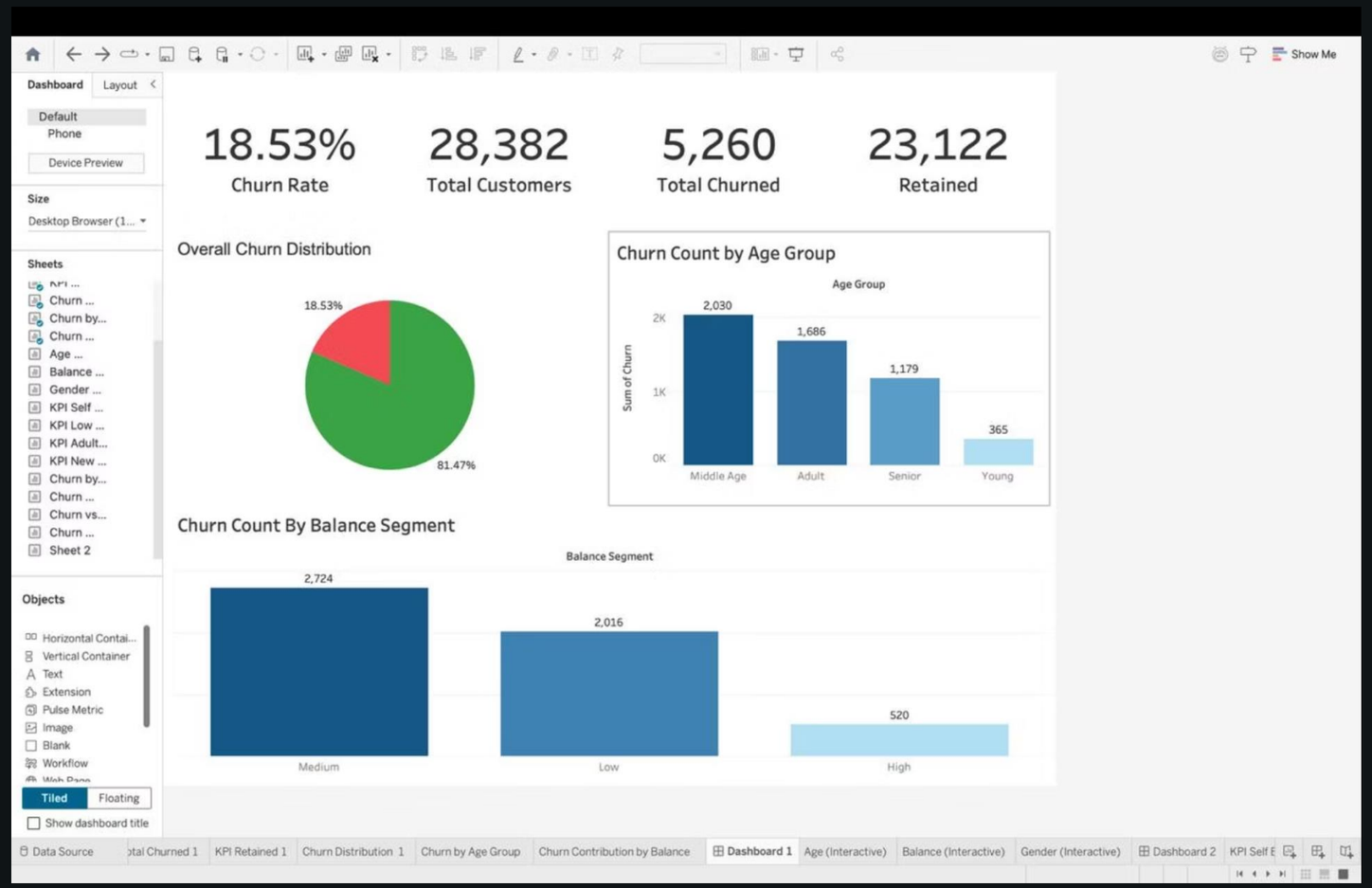
Low Activity

Lowest churn — far below average

*Segmentation also performed across **Age Groups** and **Balance Segments** to pinpoint high-risk cohorts.*

DASHBOARD OVERVIEW — SLIDE 7

Executive Dashboard: Full Churn View



📊 *Executive KPIs — 18.53% churn rate, 28,382 customers, 5,260 churned, 23,122 retained at a glance.*

📈 *Operational Charts — Churn by Age Group, Churn by Balance Segment, and Overall Distribution pie chart.*

🎛 *Interactive Filters — Slice by Age, Balance Segment, and Activity Level for targeted exploration.*

TOP INSIGHTS

What the Data Is Really Telling Us

⚡ Engaged = At Risk

*Highly active customers carry the **highest churn risk** — defying conventional retention logic.*

📦 Medium Segments Dominate

*Mid-range balance and age groups contribute the **majority of churn volume** despite not being extreme outliers.*

〰 Non-Linear Behavior

*Customer churn does not follow a simple linear pattern — **standard thresholds fail** to capture true risk.*

🚫 Engagement ≠ Loyalty

*Frequency of interaction alone cannot be used as a **proxy for retention** — quality of experience matters more.*

These insights challenge standard retention models and call for a behaviorally-informed, segment-specific approach to churn prevention.

RECOMMENDATIONS

From Insights to Action

01

Investigate High-Activity Churn

*Conduct qualitative research (surveys, NPS analysis) to uncover **why engaged customers leave** — product gaps, service failures, or competitive offers.*

02

Upgrade Service for Engaged Users

*Prioritize service quality improvements for high and very-high activity cohorts — these customers **expect more** because they use more.*

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Personalized Retention Strategies

Design segment-specific offers — loyalty rewards for high-activity users, re-engagement campaigns for mid-balance customers.

04

Early Churn Detection System

*Deploy a real-time risk scoring model to flag at-risk customers **before** they initiate churn, enabling proactive outreach.*

Impact & Expected Outcomes

BUSINESS IMPACT

📉 Reduce Churn Rate

Targeted interventions can measurably lower the current 18.53% churn baseline.

🔒 Improve Retention

Smarter, segment-aware strategies convert at-risk customers into loyal ones.

💰 Increase LTV

Every churned customer prevented directly increases customer lifetime value.

EXPECTED OUTCOMES

Predictive Accuracy

Expected churn probability accuracy of 75-85%.

High-Risk Segments

Clear identification of customer groups most likely to churn.

Measurable Churn Reduction

Quantifiable reduction in churn rate within 6 months.

PRIORITY & FEASIBILITY

High Priority

Critical to revenue protection.

Medium Feasibility

Requires ML expertise and real-time infrastructure.

ROI

Every 1% churn reduction = $2.5M+ annual revenue impact.

LIMITATIONS & CONSTRAINTS

Limitations & Constraints

Dataset lacks behavioral and transactional depth

The available dataset primarily covers static customer attributes and churn status, without granular transactional history or detailed behavioral patterns.

No real-time or time-series data

Analysis was performed on a snapshot, meaning it does not account for dynamic changes in customer behavior or evolving market conditions over time.

Limited external factors (market, competition)

The dataset does not include external economic indicators, competitor activities, or market-specific events that could influence customer churn.

Segmentation assumptions may simplify reality

The defined age groups and balance segments are based on fixed thresholds, which might oversimplify the complex, continuous nature of customer demographics and financial situations.

STRATEGIC DIRECTION

Slide 12 - Next Steps & Future Extensions

Next Steps

01

Build ML-based churn prediction model

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Integrate real-time monitoring dashboards

03

Include customer behavior & transaction data

04

Deploy targeted retention campaigns

Future Extensions

- *Integrate real-time transaction data streams*
- *Develop predictive churn scoring API*
- *Expand to other customer segments (SMB, corporate)*
- *Build automated retention campaign orchestration*

"Churn is not just a data problem, but a customer experience problem — and solving it requires both analytics and strategic action."